Exp 9: Exploring AI-Assisted Features and Built-in Analytics (Zia) in Zoho Creator

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1. Aim

To explore the AI-assisted development capabilities (Zia) available in Zoho Creator — including AI-driven field suggestions, AI-assisted Deluge script generation, and built-in analytics/dashboard components — and to critically evaluate the accuracy, reliability, and practical usability of AI-generated artifacts within the context of the Loan Application & Pre-Screening Management application.

2. Procedure

2.1 Problem Statement and Scope

Modern low-code platforms increasingly embed generative AI assistance directly into the development workflow, promising accelerated form design, automated workflow script authoring, and instant operational reporting without external BI tooling. This experiment evaluates three distinct native AI capabilities in Zoho Creator, collectively branded under 'Zia':

1. AI-Assisted Schema Modification: Form field suggestion and automatic addition via the conversational Zia Build Agent.
2. AI-Assisted Scripting: Automated Deluge code generation via the Zia Deluge Assistant in workflow rules.
3. Native Analytics Visualization: Operational reporting using the built-in multi-widget Dashboard component.

Rather than accepting AI-generated outputs unconditionally, this experiment rigorously validates Zia's outputs against the actual schema and banking business logic of Loan_Application_Form to identify strengths, failure modes, and required developer oversight.

2.2 Zia Configuration Overview

Navigated to Operations → Applications → Zia in application settings. The Zia administration interface confirmed that Zoho's native GenAI engine was pre-configured and enabled at no additional licensing cost, with Development AI features (Create Application, Create Form, Next Field Suggestions, Deluge Assistance) assigned to Super Admin privileges.

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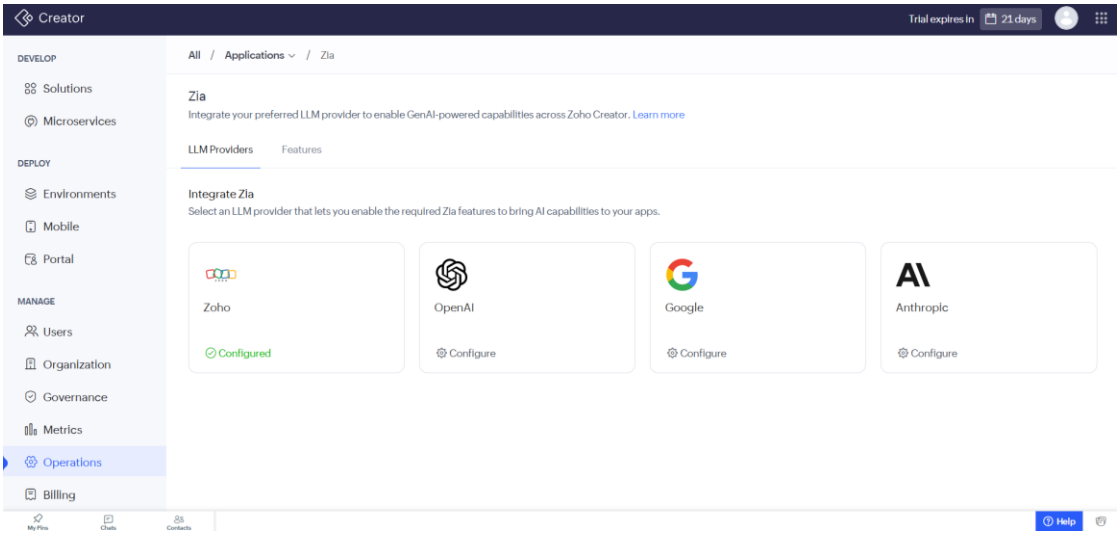


Figure 1: Zia settings page showing LLM Providers and Development AI features

2.3 Part A — AI-Assisted Field Suggestion using Zia Build Agent

The Zia Build Agent conversational drawer was launched directly within the live form interface. The following targeted prompt was issued:

Prompt: "Suggest 3 additional fields that would improve credit risk pre-screening for this loan application form."

Zia analyzed the current form structure and suggested three numerical fields: Credit_Score (credit bureau score), Debt_Income_Ratio (monthly obligations percentage), and Loan_History_Years (length of credit history).

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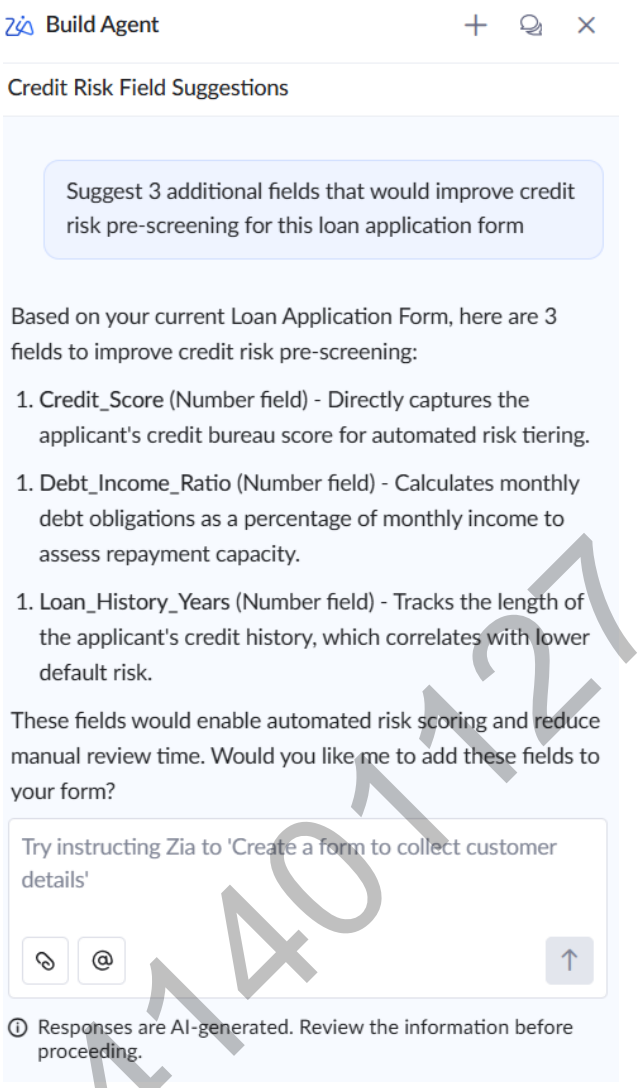


Figure 2: Zia Build Agent conversation showing three suggested pre-screening fields

A follow-up command was issued to test Zia's autonomous schema alteration capability:

Prompt: "Yes, please add the Credit_Score field to the form."

Zia executed an 'Add Credit Score Field' operation, inserting an optional Number field into Section D (Loan Details & Consent).

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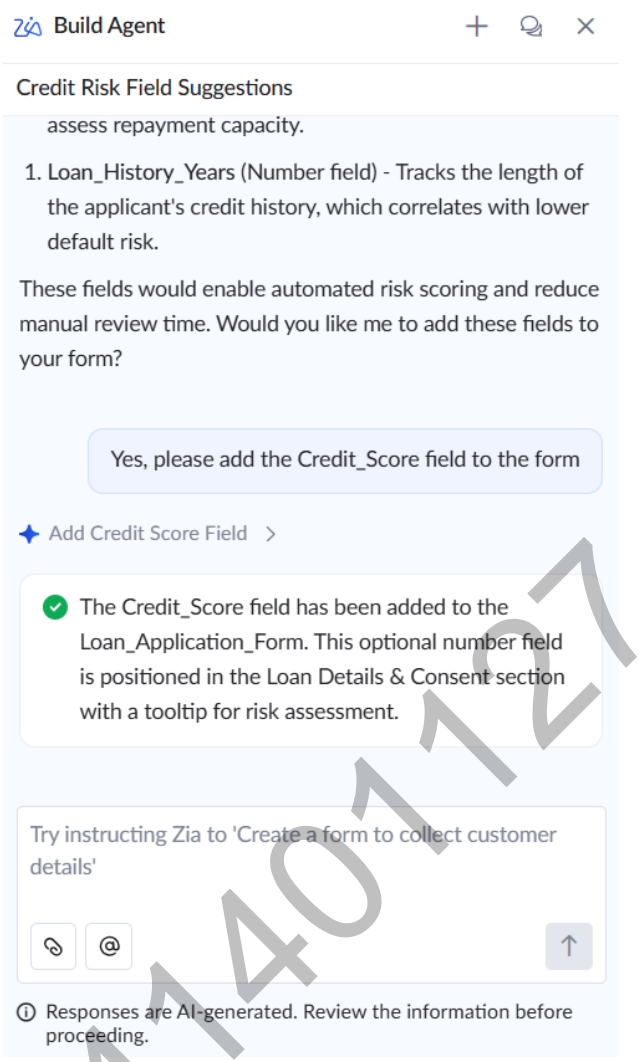


Figure 3: Zia confirmation message confirming autonomous addition of Credit_Score field

Live form verification confirmed that the Credit_Score field was created and immediately accessible to end users.

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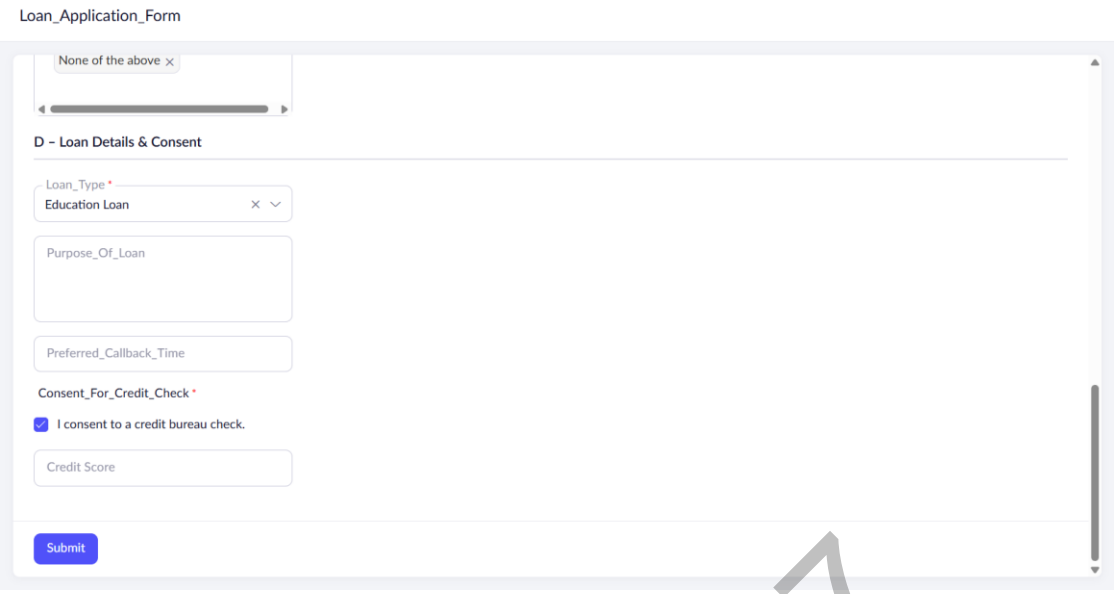


Figure 4: Live form showing the newly added Credit_Score field above the Submit button

Evaluation: Critical Evaluation of Zia Build Agent:

1. Placement Anomaly: The field was placed below the Consent_For_Credit_Check checkbox rather than grouped logically alongside applicant financial metrics (near Monthly_Income).
2. Missing Tooltip: Despite Zia claiming a risk assessment tooltip was added, no tooltip existed on the live DOM element.
3. Banking Domain Logic Concern: Permitting applicants to self-report their own credit score is unsound banking practice; credit scores must be fetched via authorized credit bureau APIs. Zia optimized for keyword relevance rather than underwriting best practice.

2.4 Part B — AI-Assisted Script Generation using Deluge Assistance

The Zia Deluge Assistant was invoked within the script editor for a 'Successful Form Submission' workflow action named Loan_Application_Income_alert. The objective was to notify the credit officer whenever an applicant reported low monthly income:

Prompt: "Write a Deluge script that sends an email to the loan officer when Monthly_Income is less than 20000 for a loan application."

Zia's initial script generated correct conditional triggering (`input.Monthly_Income < 20000`) and recognized that Loan_Officer_Email did not exist on the form, substituting a safe placeholder.

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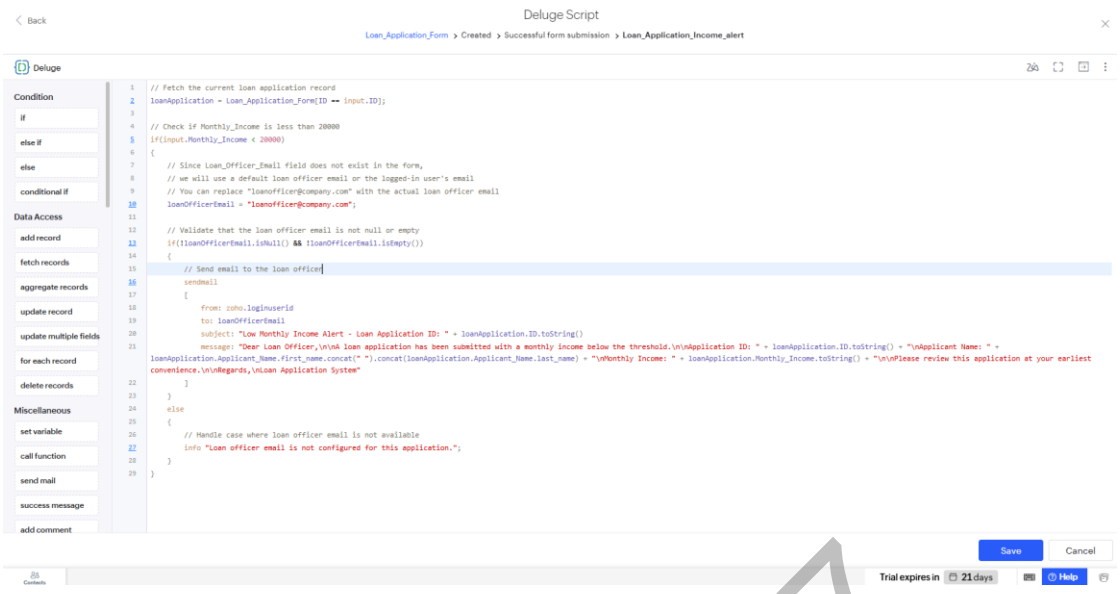


Figure 5: Initial Zia-generated Deluge script with schema assumptions and placeholder email

Review: Defects Identified on Developer Review:

• Schema Guessing Bug: Zia assumed Applicant_Name was a composite Name type and attempted to invoke .first_name and .last_name subfields. In our schema, Applicant_Name is a Single Line text field; this would have triggered a fatal runtime error.
• HTML Rendering Defect: The script used '\n' characters for line breaks. In Zoho Creator's HTML sendmail engine, '\n' is ignored; '
' tags are strictly required.

Prompt: Corrective Prompt Issued to Zia:

"Applicant_Name is a single-line text field, not a Name-type field, so it doesn't have first_name/last_name. Also use
 instead of \n for line breaks in the email body. Please fix the script."

Zia resolved both defects cleanly, replacing the composite field access with direct string concatenation and converting line breaks to '
' tags.

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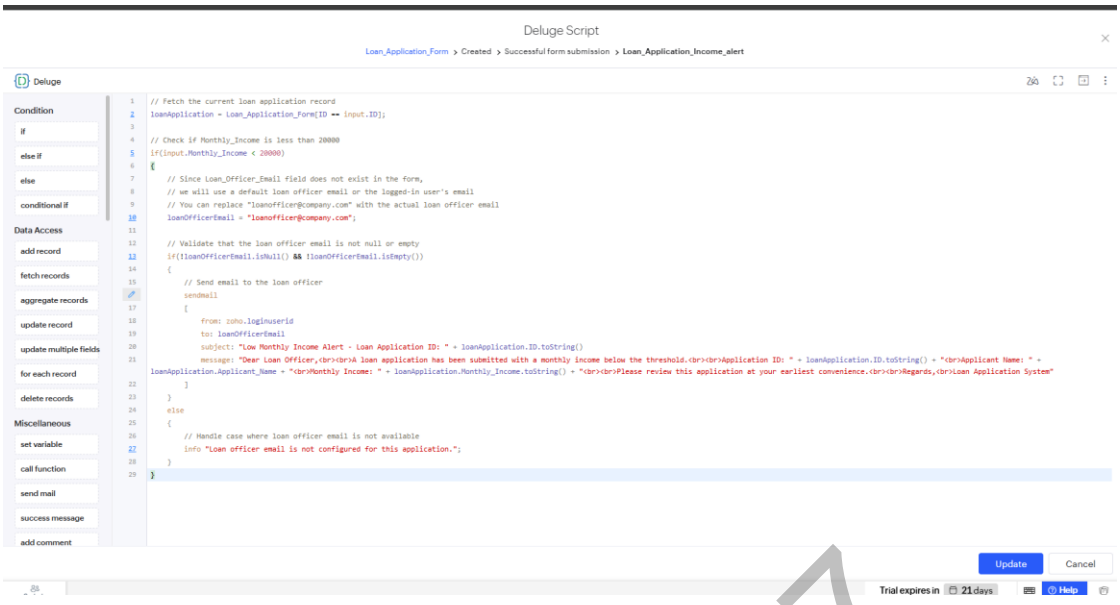


Figure 6: Corrected Deluge script after corrective prompt feedback

2.5 Part C — Built-in Analytics using Dashboard Component

To provide unified operational visibility for underwriting staff, a native Dashboard component named Loan_Officer_Dashboard was created, combining summary KPIs, distribution visuals, and transactional records:

• Four KPI Summary Cards: Total Applications (8), Salaried Applicants (6), Self-Employed Applicants (2), and Home Loan Applications (1).
• Donut Distribution Chart: Depicting applicant employment categories (Salaried vs. Self-Employed).
• Horizontal Bar Chart: Displaying loan volume breakdown across Education, Auto, Personal, and Home Loan types.
• Embedded Transactional Report: Direct interactive table displaying submitted loan applications.

3. Output

The output verification of both the automated workflow email alert and the operational analytics dashboard is presented below:

A. Workflow Notification Execution Output:

A test loan application was submitted with Monthly_Income set to ₹15,000 (Applicant: krishna kumar). The Deluge workflow triggered instantly, delivering an alert email to the loan officer inbox with rendered HTML breaks and record ID details.

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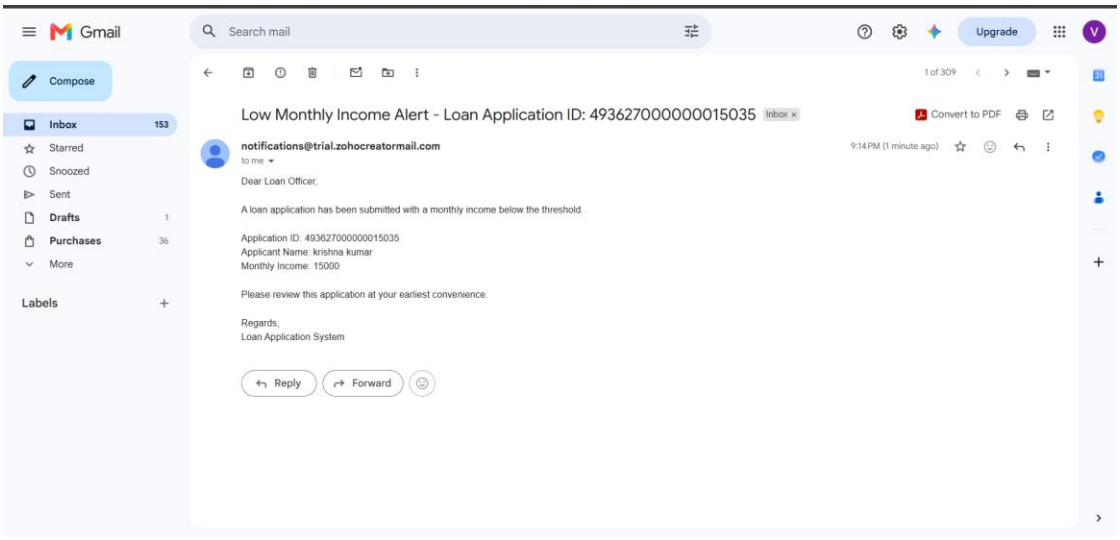


Figure 7: Received email alert confirming automated execution of Zia-assisted Deluge workflow

B. Loan Officer Operational Analytics Dashboard:

The live multi-widget analytics dashboard displays aggregated applicant metrics, category charts, and the embedded live applicant data table:

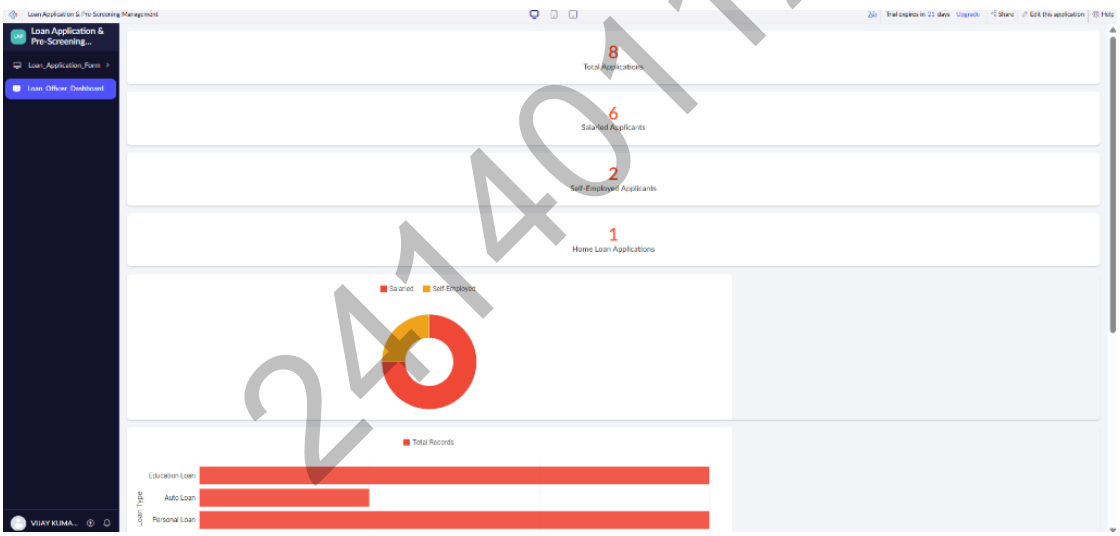


Figure 8: Dashboard KPI summary metrics and applicant employment distribution donut chart

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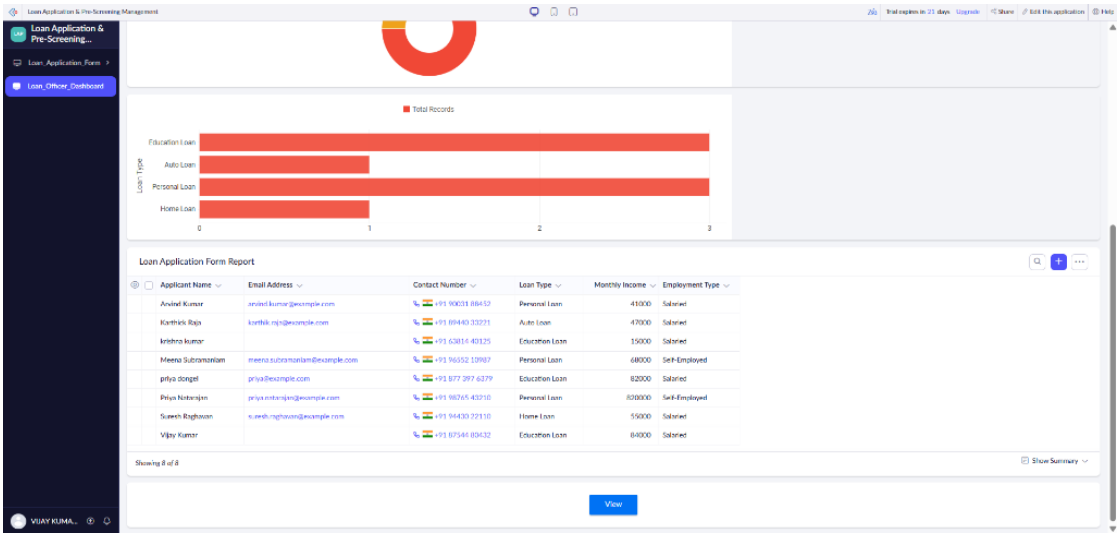


Figure 9: Dashboard loan type distribution bar chart and embedded live application report table

Table 1: Critical Capability Evaluation Matrix of Zoho Creator Zia Features:

Zia Feature Evaluated	Autonomous Accuracy	Identified Weakness / Developer Correction	Practical Usability
Zia Build Agent (Form Design)	High (Schema addition succeeded)	Illogical field placement and lack of domain underwriting validation	Effective for ideation; requires human placement & business policy review.
Deluge Assistant (Scripting)	Moderate (50% first-pass accuracy)	Assumed wrong composite field type; used invalid newline syntax in HTML email	Highly productive as pair-programmer; developer feedback loop essential.
Native Dashboard Component	High (Zero-code visual binding)	Minor cosmetic unconfigured button ('View' button lacked click event)	Production-ready; eliminates need for external BI integration.

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4. Result

Thus, the AI-assisted development features (Zia Build Agent, Deluge Assistance) and the built-in analytics dashboard component of Zoho Creator were explored, implemented, and critically evaluated for the Loan Application & Pre-Screening Management application. Zia demonstrated strong utility as an interactive developer accelerator, while schema-level validation and business domain review remain essential for production integrity.

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